

EXOCYTOSIS

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Exocytosis is a universal cellular process operative in plants, protozoa, invertebrates and vertebrates alike. It is essential for cellular differentiation and function, and intimately dependent on molecular constitution and cytomorphology. Originally, it referred narrowly to extracellular release of cell products and was termed *secretion*, *eccytosis* or *emiocytosis*. Commonly, secretion products are confined intracellularly within fluid or storage granule vesicles that fuse with the plasmalemma, and the term *exocytosis* can refer strictly to that ultimate vesicular extrusion step, or can be broadened to embody the entire functional anterograde path from site of synthesis to plasmalemmal fusion, irrespective of vesicular content(s) or membrane composition and whether granular products are released extracellularly in addition to membrane insertion into the plasmalemma.

Historical background

Cellular engulfment of solid particles via vesicles (phagocytosis) was first reported in the late 1800s. Vesicular fluid intake (pinocytosis) was described subsequently. These activities implied a large consumption of cell surface membrane, but biochemical studies in the 1950s failed to demonstrate commensurate lipid biosynthesis. This led to the concept of membrane recycling linking *endocytosis* (both phago- and pinocytosis) to *exocytosis*, respectively vesicular uptake or discharge of matter, whereby membrane segments are concomitantly respectively withdrawn from or added to the plasmalemma. Anterograde membrane flow (from cell center to plasmalemma) can originate from various organelles but ultimately homes in at specific domains of the cell surface. Membrane flow serves several functions and is involved in 1) membrane conservation, 2) endocytosis, 3) discharge from the cell of synthesized, metabolized or unmetabolizable materials, 4) transcellular transport, and 5) withdrawal from, or insertion into, the plasmalemma of lipids and integral membrane proteins, a quasisecretion process *didactically designated in this overview* as 'm(membrane)-exocytosis' to distinguish it from 's(cretion)-exocytosis', i.e. secretion/extrusion of vesicular serous or granular contents into the extracellular space. 'Exocytosis' normally refers to both of these variants. Although s-exocytosis can not occur independently of m-exocytosis, the converse can. The

term 'secretion' is more general; it does not necessarily involve vesicle-membrane fusion and presents semantic conundrums ascribable to different vocabularies of histophysiology and molecular biology; it includes 1) discharge of proenzymes from pancreatic acinar cells (merocrine secretion), 2) release of serous mixtures from sweat glands (eccrine secretion), 3) shedding of lipid from active mammary alveoli (apocrine secretion), and 4) spilling of contents upon cell death in sebaceous glands (holocrine secretion), but excludes kindred processes euphemistically called 1) 'externalization' of product-laden membrane evaginations (matrix components of cartilage and bone, some viruses, myoblast lectin), 2) 'expulsion' of nuclei from erythroblasts, or 3) 'shedding' of platelets by megakaryocytes.

Cytomorphologic infrastructure of exocytosis

Eukaryotic cells feature various membrane-delimited sacs embedded in a structured cytoplasmic matrix. These form a coherent set interconnected morphologically or via vesicular transport, thereby establishing a functionally continuous vectorial *cisternal space* (Figure 1), embedded within but topologically distinct from the cytoplasm, 'outside' the cell much as our digestive tract is topologically 'outside' our body. The cisternal space includes 1) the nuclear envelope, a perforated double-membraned sac continuous with 2) the endoplasmic reticulum, partly studded with ribosomes proximally (rough ER, RER), but not so distally (smooth ER, SER), 3) the Golgi apparatus, a set of flattened sacs shaped like soup plates nested within each other that, among other functions, encapsulates cell products into 4) membranous intracellular transport vesicles. Free cytosolic polyribosomes synthesize constitutional proteins (cytoskeletal elements, cytoplasmic enzymes, dedicated proteins such as hemoglobin). Proteins destined 'for export', i.e. m- and s-exocytosis, are synthesized by polyribosomes attached to the RER and are either anchored within the RER membrane or extruded *in toto* into the cisternal space, whence they are translocated through the Golgi apparatus for post-translational modification and/or packaging, then sorted in the trans-Golgi network into either exocytotic vesicles or lysosomes. Exocytotic vesicles with or without cargos of

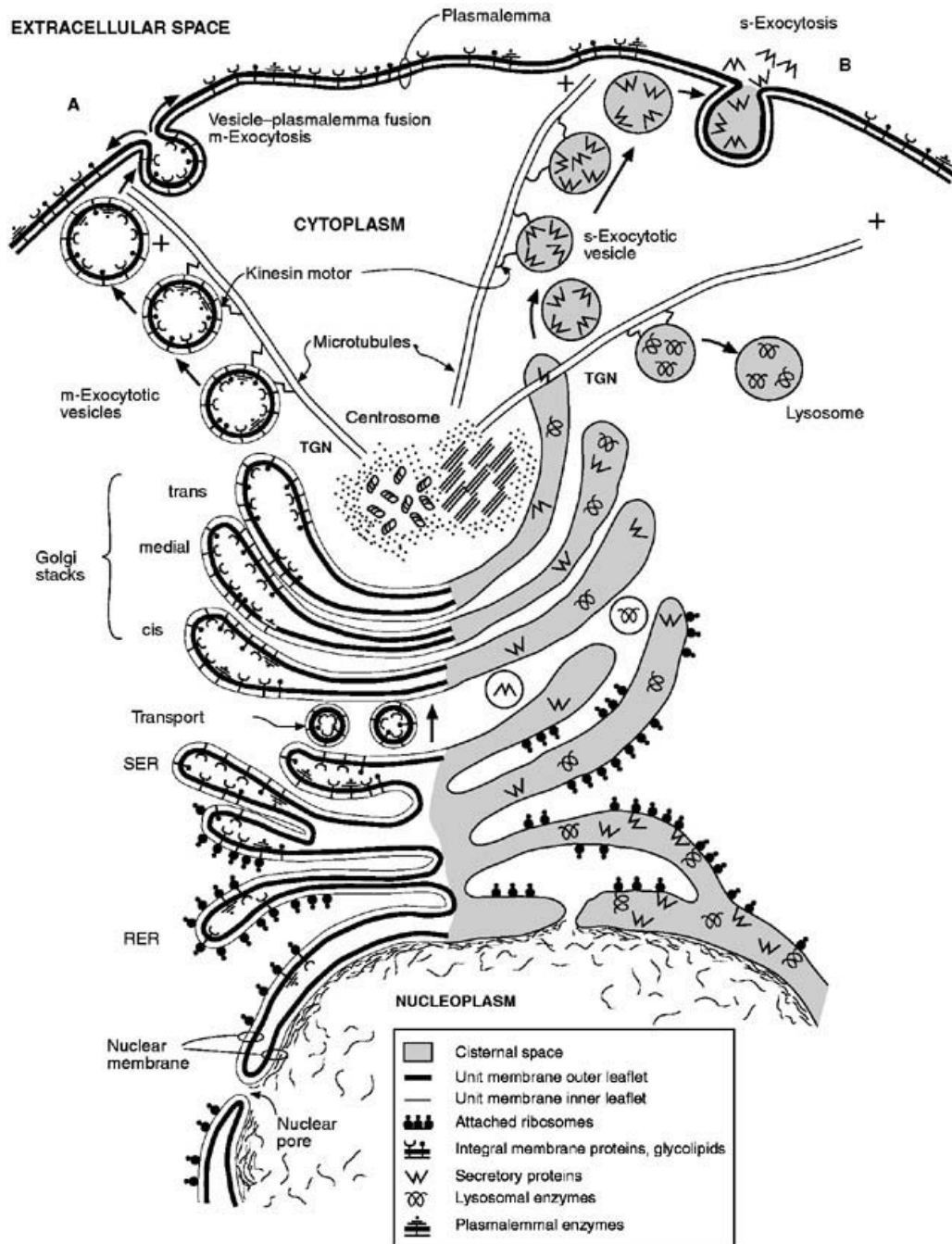


Figure 1 Exocytotic morphologic pathways for A, m-exocytotic insertion of integral membrane proteins (synaptic choline esterase, insulin or cholesterol receptors, cell attachments) and lipids (cholesterol, glyco- and phospholipids), or B, export/secretion (s-exocytosis) outside the cell. Pathway A is an obligate component of pathway B, but not vice versa. The left side of the diagram emphasizes consistency of polarity between inner and outer membrane leaflets, the right side continuity of the cisternal spaces. Integral membrane proteins (receptors, enzymes, ion gates, active transport) and glycolipids remain anchored within the lipid membranes (left side of diagram). The ultimate destination of most such proteins is the plasmalemma, although some associate with Golgi stacks, lysosomes, peroxisomes, etc. By contrast, proteins destined for release into the extracellular compartment are first injected *in toto* into the RER cisternal space and are ultimately conveyed, via contiguous segments of the cisternal space, to s-exocytotic vesicles, some propelled by cytoplasmic kinesin motor proteins to the plasmalemma along microtubules originating from the centrosome. When s-exocytotic vesicles/granules fuse with the cell surface, their membranes insert into the plasmalemma and their contents spill passively out of the cell. TGN, trans-Golgi network; RER, rough endoplasmic reticulum; SER, smooth endoplasmic reticulum; +, growing ends of microtubules. Note that s- and m-endocytosis are one and the same process that here has been dissociated for didactic purposes to emphasize the overlapping though different end-points.

specific secretory products (peptide hormones, proenzymes, etc.) first bud off from the ER, join and traverse the Golgi, and bud off again from the trans-Golgi network. Exocytotic vesicular traffic within the cell is either undirected, with docking determined by chance contacts between associated complementary targeting and accepting molecules (Table 1) on vesicular and plasmalemmal membranes, or directed to appropriate cell surface domains by cytoskeletal fibers (microtubules, actin and intermediate filaments, each associated with helper proteins modulated by specific biochemical reactions (Figure 2). s-Exocytosis occurs 1) of zymogen granules by gastric chief cells at the apical surface, 2) of chylomicra by enterocytes, at the lateral membrane behind the junctional complexes that anchor epithelial cells to each other, restrict membrane flow between apical and basolateral domains, and insulate tissue spaces from lumina, and 3) of plasma components for transfer into tissues across endothelia, at the basal cell surface. Endocytic vacuoles can be translocated to primary lysosomes (Figure 3) where fusion of the two yields secondary lysosomes, thence back to the cell surface, a path characteristic of receptor-mediated phagocytosis; their receptors can be recycled. Polarized motor proteins, cytoplasmic dyneins and kinesins, translocate some vesicles along microtubules toward or away from the cell center, respectively, but other mechanisms may exist. Interference with microtubule assembly by colchicine impedes secretion of pancreatic zymogens, parathormone and fibroblast procollagen.

s-Exocytosis to the extracellular space

s-Exocytosis can be *regulated* or *constitutive*. Regulated exocytosis occurs on demand via secretagogues and signal transduction. Constitutive exocytosis is continuous; no secretagogue is involved, there are no storage vesicles, and the products are released apace upon synthesis. s-Exocytosed products of nonneuronal cells (digestive proenzymes, peptide hormones, immunoglobulins, chylomicra lipoproteins) originate in the RER, are carried by vesicles to and through the Golgi apparatus, then by vesicles or membrane-bounded granules to the plasmalemma, and are finally extruded by fusion of the vesicles with the cell membrane. For that entire transit they remain confined topologically within the cisternal space (Figure 1). In neurons some vesicles recycle and reload near the synapse. In pancreatic acini, fusion of progressively deeper granules with more peripheral ones, called *sequential* or *compound exocytosis*, forms extended extrusion channels, also said to occur in mast cells. Various agents elicit release of mast

cell mediators: Ca^{2+} ionophores, cross-linkers of immunoglobulin E (IgE) receptors on the cell surface by antigen, polycations or some lectins. Phagocytosis of negatively-charged thorium dioxide particles or plastic beads by rat serosal mast cells delivers them over time via endosomes to young, then mature granules, without causing degranulation. The metal is retained for months and migrates between granules, which change their staining properties as the cell ages. They have *in situ* enzymatic activity (and *ipso facto* function?), and their mediators have long half-lives. Polylysine causes degranulation. Cationic (polylysine-coated) beads breach the granular sac by a different path that bypasses endosome vesicles, and they contact mature granules directly without delay. All this argues for a transient or permanent membranous sac enfolding multiple granules in a way unrelated, if morphologically resembling, sequential exocytosis. Could IgE antigens act as *pathologic* mast cell secretagogues? This paradoxical dichotomy suggests that mast cell degranulation interferes with a primary intracellular function, whatever that might be, and that evidence for compound exocytosis must be interpreted cautiously.

At plasmalemmal docking sites, a series of biochemical signaling reactions involving G protein(s), enzymes, phosphoinositol metabolites, etc. may lead to transient shifts of ionic calcium into cells and/or between some intracellular compartments. These Ca^{2+} shifts may not be obligate steps for all cells or all s-exocytotic processes. As fusion of a vesicle with the cell membrane establishes continuity between vesicular contents and extracellular space, all the contents stored within are disgorged simultaneously (epinephrine, enkephalins, chromogranins, nucleotides, etc., by adrenal chromaffin cells), faster and in much higher concentrations than possible via active transport mechanisms. Vesicular pathways can couple endocytosis with s-exocytosis to transport material across a cell (endothelium, kidney tubules) by *transcytosis*. Though uptake at one surface may be selective, there need be no modification of the vesicular contents before s-exocytosis at another surface. Capillary endothelium transfers bulk plasma into tissue spaces, and selected substances across the blood-brain barrier. IgA and IgM are moved across enterocytes in the opposite direction, from basal to apical surfaces.

m-Exocytosis to the cell membrane

Membrane traffic is critical for defining cell type and function which depend on plasmalemmal

Table 1 Some identified components of the membrane fusion machinery in exocytosis and endocytosis^a

1. Vesicular coats -- shape membranes into vesicles	
Clathrin	Trimeric protein complexes that shape budding vesicles. Forms triskelion cages under coated pits, endocytic vesicles, and buds on trans-Golgi membranes
Coatomer	Coat protomer. Non-clathrin polypeptides condensations around shuttling cytoplasmic vesicles. At least seven COP components recognized; form stoichiometric complexes with ARF. Drives vesicle blistering of Golgi membranes
COP1 coats	Coat Proteins forming ARF-coatomer complexes around vesicles that shuttle between ER and cis-Golgi, and vice versa. Unstable; assembled by ARF-GTP
COP2 coats	As COP1, but stable. Assembled by ARF-GTP. Dissociated by hsp70 or SAR1 ATPases. Direct vesicles from ER to cis-Golgi
ARFs	ADP-Ribosylation Factors that regulate vesicular coat structures. These GTPases assemble with COPs to trigger formation of coatomers. Bind to Golgi membranes and cause vesicle budding or uncoating. ARFs are myristylated; they are cytosolic (water soluble) in GDP form, but membrane bound (water insoluble) in GTP form
AP1	Assembly Polypeptide. ARF-dependent complex vesicular protein coat promoting direct transport from trans-most Golgi cisterna to lysosomes. Associated with clathrin
AP2s	Family of Assembly Polypeptides. ARF-independent stable clathrin-associated protein complexes. Associated with coated pits, endocytotic vacuoles, and presynaptic plasmalemma of nerves
SAR proteins	ARF-like G-protein ATPases
Rab proteins	Family ($n = 30+$) of small (22–27 kDa) G proteins with GTPase activity; required for assembly of SNARE complexes. Associated with multiple stages of vesicular cytoplasmic traffic in exocytotic and endocytotic pathways. Regulate or trigger secretion (of enzymes, hormones, neurotransmitters, etc.) via stimulation or inhibition
Annexin XIII	Nonmembrane protein implicated in vesicle transport to apical plasmalemma
TAP (p115)	Protein needed for docking and fusion of transcytotic vesicles to apical cell membrane as well as transport of vesicles between Golgi stacks
2. Membrane targeting and docking	
v-SNAREs	Family of vesicular integral membrane coat proteins that act as destination codes with affinity for cognate SNAP receptors on target membranes. Effect docking with t-SNAREs
t-SNAREs	Family of SNAPs–NSF complexes. Integral membrane proteins that act as destination codes in receptive target membranes. Act as SNAPs receptors to effect docking of specific v-SNAREs
VAMPs	Family of Vesicle-Associated Membrane Proteins. Synaptosome-associated integral membrane protein of brain (synaptobrevin) involved in neurotransmitter release; associated with dense-core vesicles (loaded and recycled at trans-Golgi, in perikaryon) and with small vesicles (loaded and recycled at neuron terminal). Also present in gastric parietal cells, zymogen and chromaffin granules, and yeast
Syntaxins	Integral membrane proteins present in presynaptic plasmalemma, gastric parietal cells and yeasts
3. Membrane fusion	
NSFs	Family of NEM-Sensitive Fusion ATPase proteins. Complex with SNAPs and SNAREs to effect vesicle fusion. Associated with ER, Golgi, plasmalemma and endosomes.
SNAPs	Soluble NSF Attachment Proteins. Colocalize with NSF and SNAREs to elicit fusion of intracellular membranes in animals and yeasts. Associated with Golgi membranes
SNAP25	SyNaptosome-Associated Protein 25 kDa. A presynaptic peripheral membrane protein (structurally unrelated to SNAPs) involved in neurotransmitter release upon local Ca^{2+} influx into the nerve. Induces rapid axonal transport. In gastric parietal cells, elicits expression of $\text{H}^{+}/\text{K}^{+}$ -ATPase and $\text{K}^{+}/\text{Cl}^{-}$ channels on surface of extended microvilli to activate HCl secretion.
Synaptotagmins (p65)	Family of integral membrane proteins, primary Ca^{2+} receptors of presynaptic plasmalemma. Inhibit neurotransmitter release. Also implicated in recycling of dense-core neuropeptide vesicles and clathrin-coated small synaptic vesicles at nerve terminals. Interacts with AP2, syntaxin, calmodulin and phospholipids. Involved in both exocytosis and endocytosis. Synaptotagmins can effect both positive and negative regulation of neurotransmitters. Also present in fibroblasts
Synaptophysins	Family of integral membrane proteins of small synaptic vesicles. May enhance exocytosis. Nonneuronal homologs reported in fibroblasts and other cells

Table 1 Continued

4. Ancillary agents	
Nucleotides	GTP and ATP. Cytosolic energy sources for assembly and disassembly of coat proteins in conjunction with GTPases and ATPases. Induce conformational changes in specific molecules
Ca ²⁺ ion	Second messenger for many exocytotic and endocytotic processes, including release of neurotransmitters
c-AMP	Second messenger in many transduction signaling processes
Secretagogues	Primary agents (polypeptide hormones, catecholamines, etc.) that trigger exocytosis by cognate cells. Generally specific ligands for target cell receptors
Fatty acyl-CoA	Alter water/lipid solubility of G proteins and effects membrane detachment/attachment
Proton pumps	ATPases that acidify endosome-lysosome vesicles to activate lysosomal enzymes and dissociate ligands from receptors
Kinesins	Family of protein motors effecting anterograde translocation of transport vesicles along microtubules
Cytoplasmic dyneins	Family of protein motors effecting retrograde translocation of transport vesicles along microtubules
Shaperones	Proteins (e.g. hsp70) that control folding/unfolding of proteins in exocytotic processes

^aCurrent status of participating agents in intracellular vesicular membrane traffic: factual, putative and hypothetical. Entries are not necessarily operative in all cell types, and their specific properties/role(s) are not fully elucidated. The proteins involved in the highly specific and complex processes that lead to exocytosis are members of families of homologous proteins represented in all eukaryotic cells. The mechanisms of transport, docking, fusion and exocytosis are highly conserved from yeast to mammalia, albeit with some specializations and/or variations among cell types and species, with many isoforms present in given species, tissues, cells and even single vesicles. The table includes only a few of the plethora of reported agents. Many additional ones unquestionably will be characterized in this rapidly-developing research area.

differentiation and maintenance. It modulates up-/downregulation of surface receptors involved in signaling and thus actuates switches that control cell function. Cell membranes present domains of differing composition and functions; they assure epithelial polarity which, when disturbed, can lead to severe pathology. The plasmalemma is a mosaic of integral proteins floating in a bilayer of hydrophobic lipids. Intracellular membranes define various aqueous compartments within the cytoplasm. For m-exocytosis the proteins, synthesized by the RER, remain anchored in a polarized orientation within the lipid bilayer by means of hydrophobic sequences in their molecular backbone; they can diffuse laterally within the plane of the membrane, but can not flip-flop. Proteins that project into the cisternal space can be glycosylated or otherwise modified post-translationally. m-Exocytosis of vesicles budded off the trans-Golgi network is the vehicle for translocating these molecules from their site of synthesis to their site of function at the plasmalemma. It insures that their orientation in the membrane is retained during translocation to and fusion with specific domains of the cell membrane, in which they may then diffuse or aggregate laterally to form specialized functional sites (enzyme or transport proteins at the tip of enterocyte microvilli, ion channels, receptors, integrin binding sites for fibronectin) or components of cell-cell junctions (desmosomes, synapses, tight, gap and neuromuscular junctions). Lipids and glycolipids made by the ER-Golgi complex are translocated similarly. Membrane differentiation gives cells their individuality and many of their special functions. It is how cells develop functionally different apical,

lateral and basal surfaces, express specific surface molecules (ABO antigens, receptors) or recognition surface proteins (T lymphocytes).

Membrane flow and conservation

Membrane traffic is essential to the dynamic responsiveness of cells. Plasmalemmal turnover can be massive and very rapid. Because membrane augmentation by exocytosis is coupled with endocytosis, which withdraws membrane from the cell surface, an equilibrium is established and membrane components can be efficiently conserved, added, removed, replenished, changed or recycled. Extracellular ligands snagged by surface receptors can be drawn into the cytosol by endosomes that fuse with lysosomes and become acidified, thereby dissociating ligand from receptor. Dysjunction of the endosome-lysosome complex yields a digestive vesicle and a shuttling receptor-carrier vesicle that can rejoin the plasmalemma via m-exocytosis for another cycle (Figure 3). Normal topological relations are unaffected in those exchanges, as critical membrane polarity is retained (Figure 1). Lastly, massive fusion of exocytotic vesicles can effect cell shape changes (yeast shmooing (formation of pear-shaped cells during mating), microvilli extrusion in activated gastric parietal cells).

The above discussion clearly points to the central role of exocytosis in the very existence of living cells. DNA directs synthesis of cell constituents that establish cellular ultrastructure, but it is cell structure that, via exocytosis, executes the myriad processes that

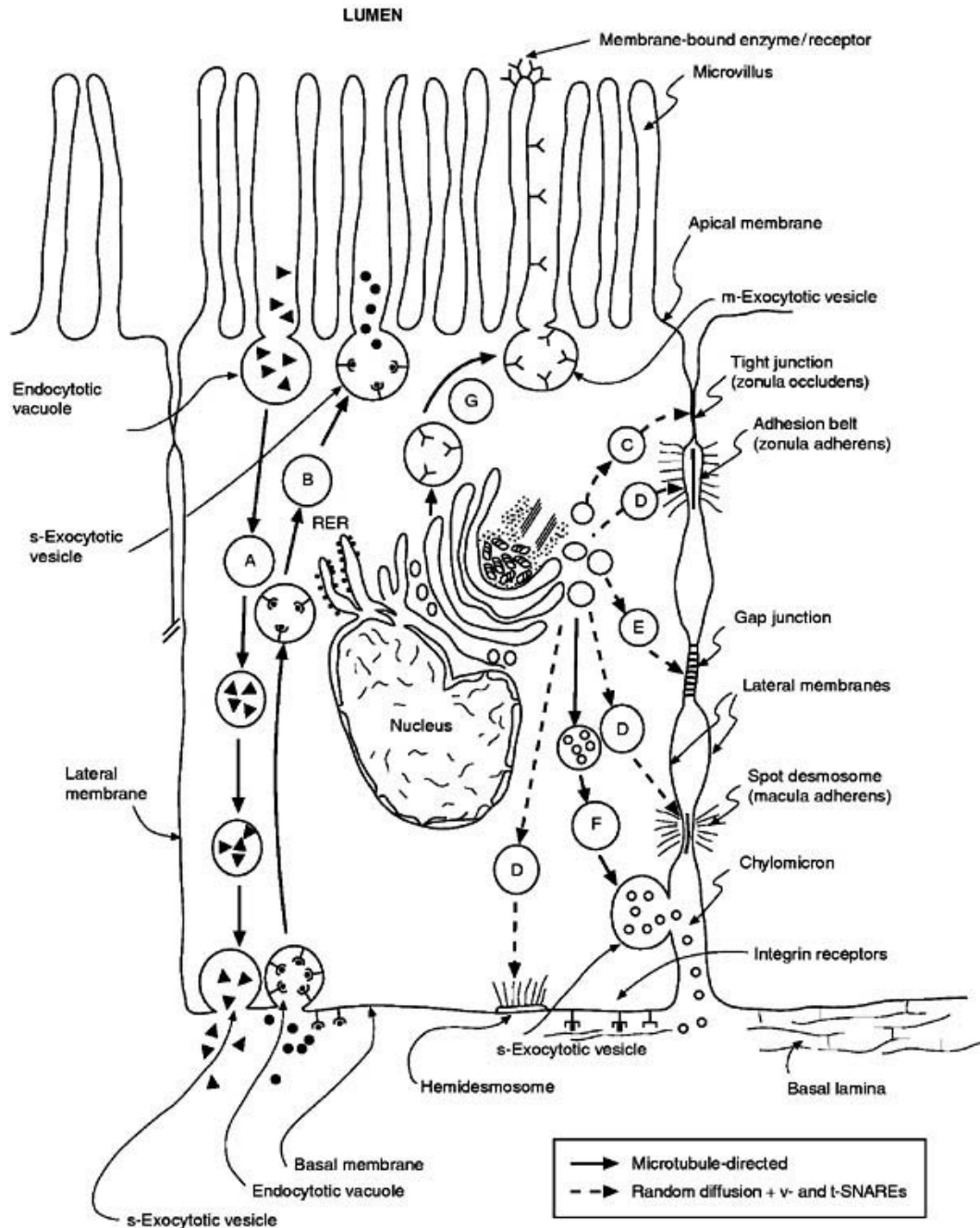


Figure 2 Targeting/capture of m-exocytotic and s-exocytotic vesicles to/by specialized cell membrane domains. A polarized epithelial cell is depicted, but many of the same features are found in other cell types (neurons, macrophages, lymphocytes, yeast). A, Transfer of substances across an epithelial cell (endothelium, kidney tubule) via a transcellular path from apical (endocytosis) to basal (s-exocytosis) domains; B, uptake of extracellular material (endocytosis) for transfer to luminal space (s-exocytosis) from basal to luminal domains (IgA, IgM across enterocyte); C, m-exocytosis of integral membrane protein (uncharacterized) to tight junction between lateral domains of abutting epithelial cell membranes; D, m-exocytosis of adhesion integral membrane proteins (CAMs, integrins) to form spot and belt desmosomes and hemidesmosomes at lateral and basal membrane domains; E, m-exocytosis of integral membrane proteins (connexin) to communicating (electrotonic) gap junctions in the lateral domain of abutting cells; F, m-exocytosis of receptors to basal membrane (binding to fibronectins, glycosaminoproteoglycans, intermediate filaments). Specialized cell junctions are transient in that they can be withdrawn and restored rapidly under appropriate conditions. The bimolecular unit membranes are drawn as a single line. Note that s- and m-endocytosis are one and the same process that here has been dissociated for didactic purposes to emphasize the overlapping though different end-points.

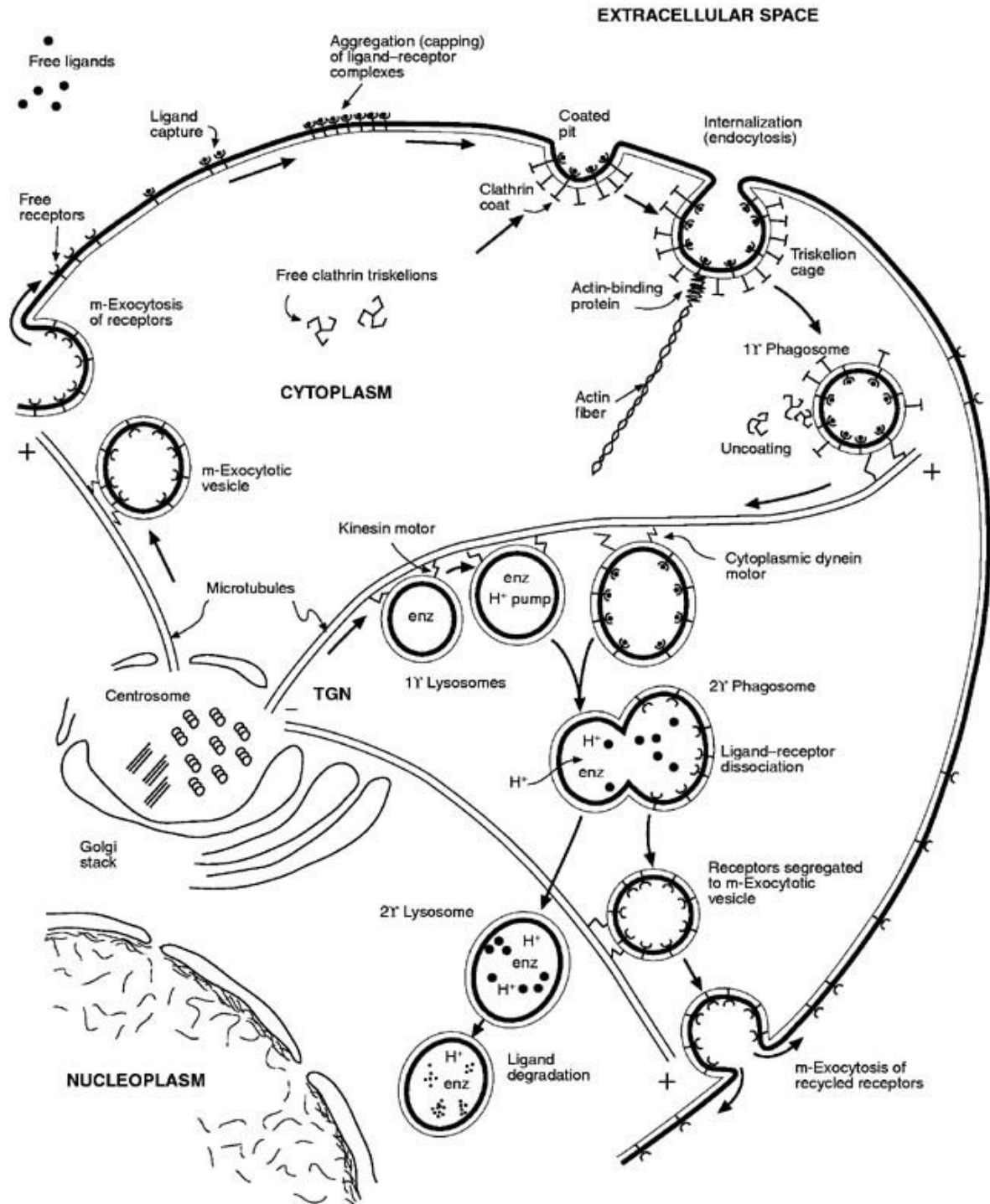


Figure 3 Endocytosis–exocytosis coupling, membrane conservation and receptor recycling. Integral membrane proteins such as receptors delivered to the plasma membrane by m-exocytosis (Figure 1) bind specific ligands, and the complexes then aggregate into clathrin triskelions-coated pits that are internalized by endocytosis of vesicles linked to and pulled off the plasmalemma by actin filaments. After internalization, the triskelion cage depolymerizes and the vesicle connects to microtubules via cytoplasmic dynein motor proteins that move it along retrogradely. The vesicles join and fuse with lysosomes shed off the trans-Golgi network and propelled anterogradely by kinesin motor proteins, also along microtubules. The fused vesicles become acidified by an ATP-driven proton pump, the receptor–ligand complex dissociates and the lysosome–vesicle organelle (secondary lysosome) reseparates into lysosome and vesicle moieties. The freed ligand, sequestered within the secondary lysosome, is degraded by acid hydrolases, while the receptors sequestered within the regenerated m-exocytotic vesicle membrane are translocated back to the plasmalemma for recycling. TGN, trans-Golgi network complex; H⁺, proton; enz, lysosomal enzymes; triskelions, clathrin trimers. Note that s- and m-endocytosis are one and the same process that here has been dissociated for didactic purposes to emphasize the overlapping though different end-points.

breathe life into the organism and assure survival of its DNA.

See also: Adhesion molecules; Antigens, cell surface; Atopic allergy; Capping, clustering, membrane microdomains and all surface dynamics; CD antigens; Cell-mediated immunity; Complement receptors; Degranulation; Fc receptors; Hypersensitivity reactions; Immunoglobulin, cell surface; Mast cells; Phagocytosis.

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EXPERIMENTAL AUTOIMMUNE ENCEPHALOMYELITIS

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Experimental autoimmune encephalomyelitis (EAE) is an inflammatory disease of the central nervous system (CNS) which can be induced in susceptible strains of laboratory animals by immunization with CNS antigens. It is characterized clinically by ascending paralysis, and pathologically by perivascular mononuclear cell infiltration and demyelination in the brain and spinal cord. Although EAE is an experimental disease of laboratory animals, it was first observed in humans when, over a century ago,

it was reported that a small percentage of patients receiving Pasteur's rabies vaccine developed neuro-paralytic signs. The rabies virus vaccine was prepared in rabbit CNS tissue. That CNS antigens contaminating the vaccine preparations were probably responsible for the neurological side-effects of the vaccine was demonstrated in the 1930s, by Rivers and his colleagues, who reported that repeated intramuscular injection of extracts of normal rabbit brain (i.e. without rabies virus) induced pathologic changes